

- The cochlear division is formed of central processes of the **spiral ganglion** that receives signals from the **organ of Corti** (hearing receptor).
- The vestibular division is formed of central processes of the **vestibular ganglion** that receives from the **3 semicircular canals & vestibule** which mediate balance and motion.
- At the **internal acoustic meatus**, both parts unite to form the vestibulocochlear nerve.
- The nerve attaches to the lower border of **pons** lateral to facial nerve in the cerebello-pontine angle.

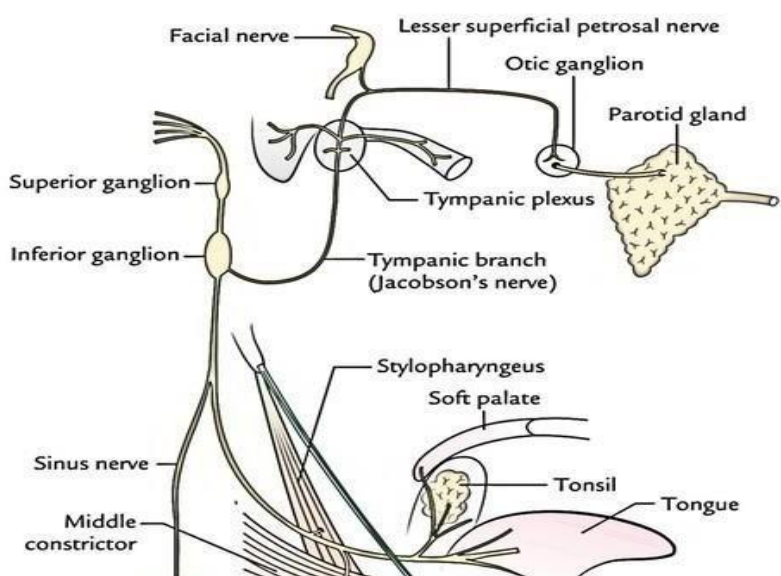
Glossopharyngeal nerve (CN IX)

- **Origin:** it is a mixed nerve that arises from 4 nuclei that are in the medulla oblongata:
 - 1- Nucleus ambiguus (motor).
 - 2- Inferior salivary nucleus (parasympathetic).
 - 3- Spinal nucleus of trigeminal (general sensation).
 - 4- Solitary nucleus (taste).
- **Course:**
 - The nerve emerges from the posterolateral sulcus of **medulla**.
 - It leaves the cranial cavity through the **jugular foramen**.
 - It descends in carotid sheath. Then, it passes between **ICA & ECA**.
 - It enters the pharynx between the **superior & middle constrictors**.
- **End:**

It terminates by dividing into many branches supplying the posterior third of tongue.

▪ **Branches:**

- 1- **Tympanic branch:** carries parasympathetic fibers to



parotid.

2- **Carotid branch** to carotid body.

3- Nerve to stylopharyngeus.

4- **Pharyngeal sensory branches.**

5- **Tonsillar sensory branch** to the palatine tonsils

6- **Terminal lingual branches:** sensory to the posterior 1/3 of the tongue.

Vagus nerve (CN X):

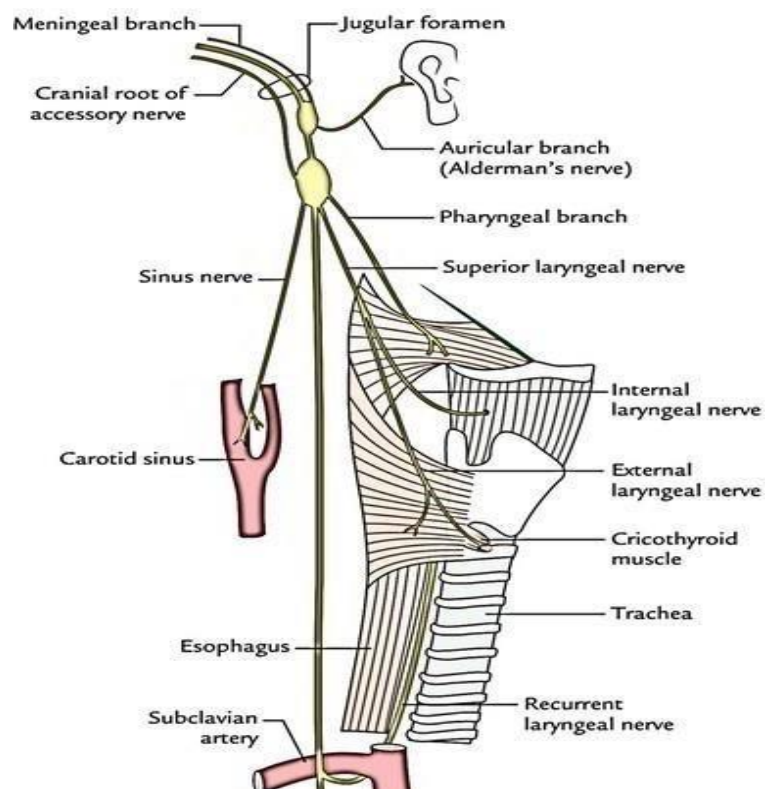
- **Origin:** The **vagus nerve** is a mixed nerve that arises from 4 nuclei that are in the medulla oblongata:
 - Nucleus ambiguus (motor).
 - Dorsal motor nucleus of vagus (parasympathetic).
 - Spinal nucleus of trigeminal (general sensation).
 - Solitary nucleus (taste).

Course:

- The nerve emerges from the posterolateral sulcus of **medulla**.
- It leaves the cranial cavity through the **jugular foramen**.
- It descends between IJV & ICA in carotid sheath.
- It enters the root of neck in front of **the 1st part subclavian artery**.

▪ **Branches:**

- 1- **Meningeal branch** to the dura of the posterior cranial fossa.
- 2- **The auricular branch** to the



skin of external
auditory meatus.

3- **Carotid branch:** to carotid
body.

4- **Pharyngeal motor branches:** to the palatal muscles except the tensor palati &
pharyngeal muscles except the stylopharyngeus.

5- **Superior laryngeal nerve:** for sensory supply of the mucus membrane of larynx
above the level of vocal cords & motor to cricothyroid.

6- **Recurrent laryngeal nerve:** for sensory supply of the mucus membrane of larynx
below the level of vocal cords & motor to muscles of larynx except cricothyroid

7- **Cardiac branches:** parasympathetic to the heart.

Accessory nerve (CN XI)

▪ This **nerve** is formed by the union of cranial and accessory roots at jugular foramen.

▪ **Origin:**

○ The **Cranial root** emerges from the posterolateral sulcus of **medulla**. It
originates from motor **nucleus ambiguus** in medulla.

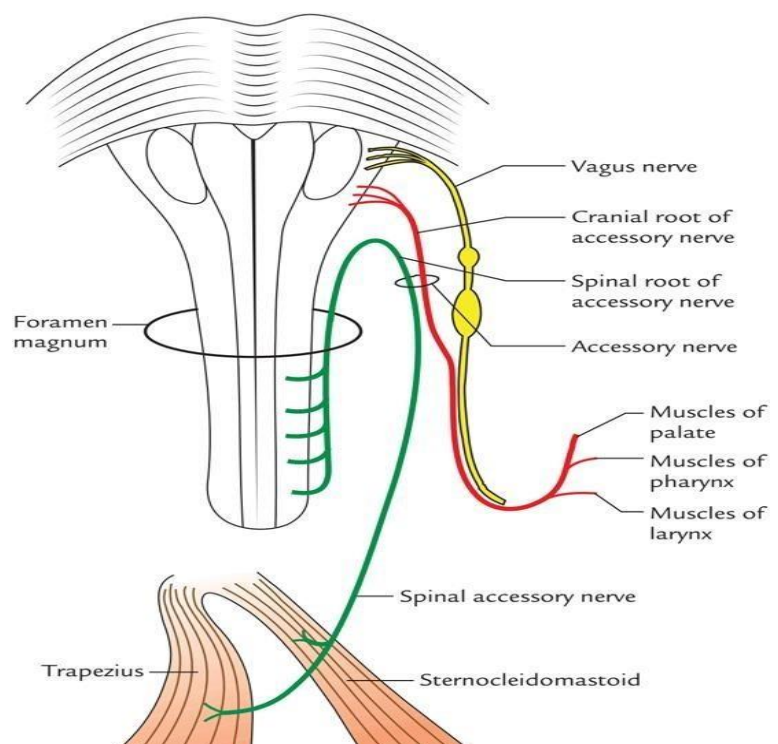
○ The **Spinal root:** arises from AHCs of **upper 5 cervical segments**.

▪ **Course:**

○ The **spinal root** enters the
cranial cavity through the
foramen magnum to join
the cranial root at jugular
foramen.

○ Just outside the jugular
foramen the two
roots separate again.

○ The cranial part joins vagus
and distributes with its
branches.



- **Branches:**

- The **cranial accessory** supplies through vagus:
 - 1- Palatal muscles except tensor palati.
 - 2- Pharyngeal muscles, except the stylopharyngeus.
 - 3- Laryngeal muscles.
- The spinal root reaches the posterior triangle of neck to supply: **Sternomastoid & trapezius muscles.**

- **Lesion** to spinal accessory in posterior triangle of the neck leads to **drooping of the shoulder** due to paralysis of **trapezius muscle** only.

Hypoglossal nerve (CN XII):

- **Origin:**

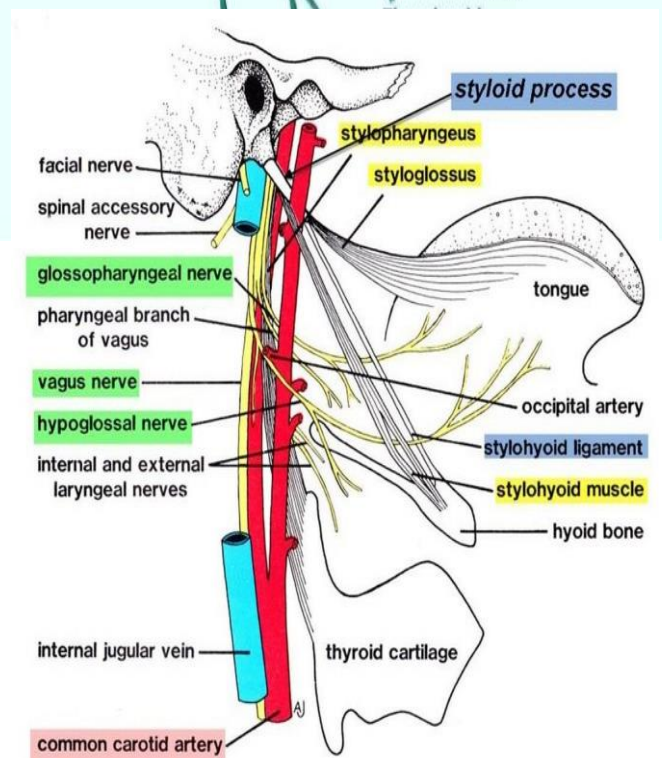
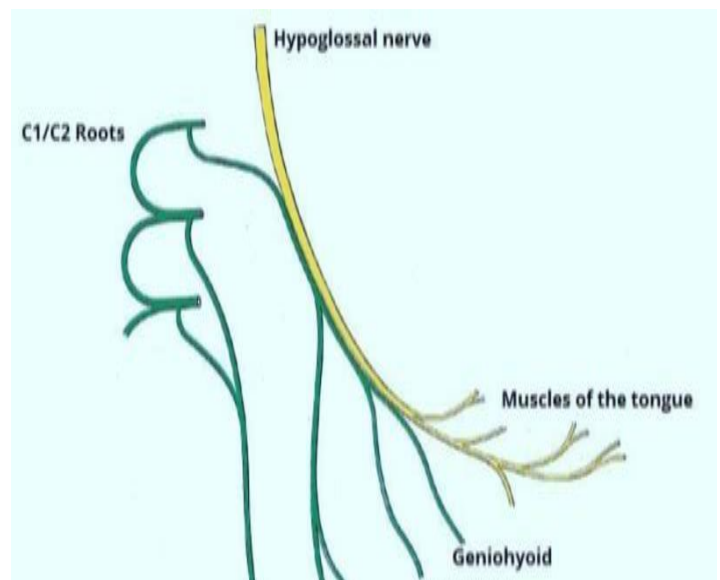
- This nerve emerges from the **medulla** between the pyramid & olive.
- It arises from hypoglossal motor nucleus in medulla.

- **Course:**

- The nerve leaves the cranial cavity through the anterior condylar canal. It enters the carotid sheath.
- It passes forward between ICA & IJV, the superficial to ECA.
- It enters the tongue superficial to hyoglossus muscle.

- **Supply:**

- **All tongue muscles** except the palatoglossus which is supplied by the cranial root of accessory.
- The joining fibers of **C1** supply:



- 1- **Descendens hyoglossi** of ansa cervicalis.
- 2- **Nerve to thyrohyoid & geniohyoid** muscles.
- **Lesion of the nerve** leads to deviation of the tongue towards the paralyzed side when protruding the tongue.

CERVICAL PLEXUS:

- **origin:** formed by the ventral primary rami of the **upper 4 cervical nerves**.
- **Site:** on the anterior surface of the Scalenus Medius.
- **Branches:**
 - **Motor branches:**
 - 1- Phrenic nerve: from C 3, 4, 5.
 - 2- Descendens cervicalis: from C 2, 3.
 - 3- To trapezius & levator scapulae: from C 3, 4.
 - 4- To sternomastoid muscle: from C 2, 3.
 - 5- To Scalene muscles.
 - **Sensory branches:**
 - 1- **Lesser occipital nerve:** supplies skin behind the auricle.
 - 2- **Great auricular nerve:** supplies skin over the angle of mandible.
 - 3- **Transverse coetaneous of the neck:** supplies skin of the side & the front of the neck.
 - 4- **Supraclavicular nerves** (medial, intermediate & lateral) supply the skin of shoulder region.

supply the submandibular gland & also **taste fibers** to the anterior 2/3 of the tongue.

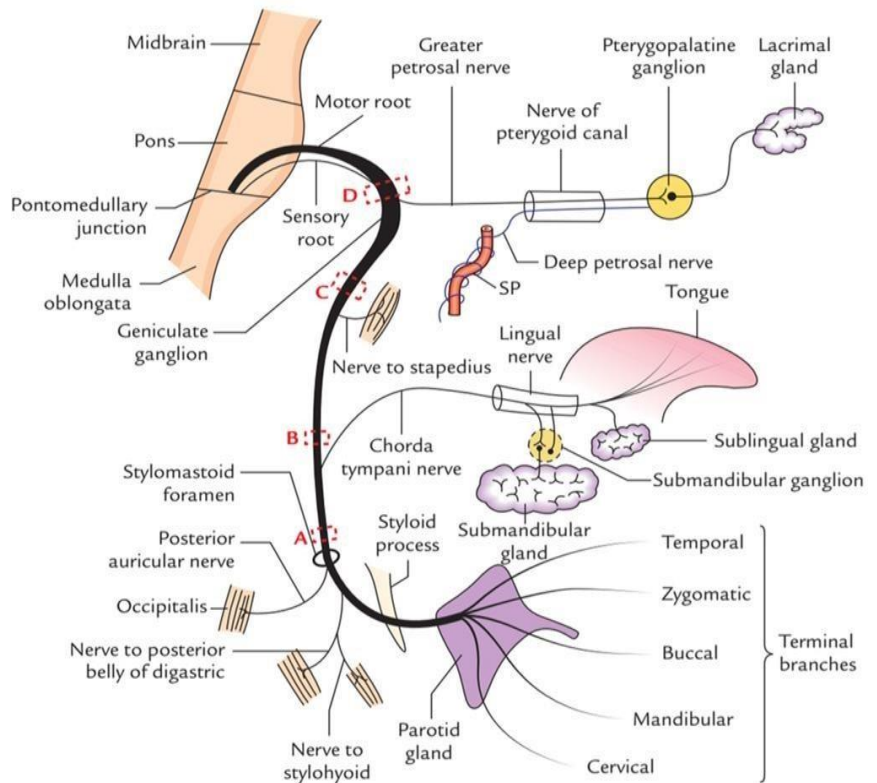
○ **Extracranial part:**

- Just below stylomastoid foramen, it gives off:

- 1- **posterior auricular nerve** supplies occipitalis muscle.
- 2- **Nerve to posterior belly of digastric and stylohyoid muscle.**

- It gives **five terminal** branches within the parotid gland:

- 1- **Temporal nerve** supplies frontalis and orbicularis oculi.
- 2- **Zygomatic nerve** supplies orbicularis oculi muscle.
- 3- **Buccal nerve** supplies buccinator & muscles of upper lip.
- 4- **Marginal mandibular nerve** supplies muscles of lower lip.
- 5- **Cervical nerve** supplies platysma.

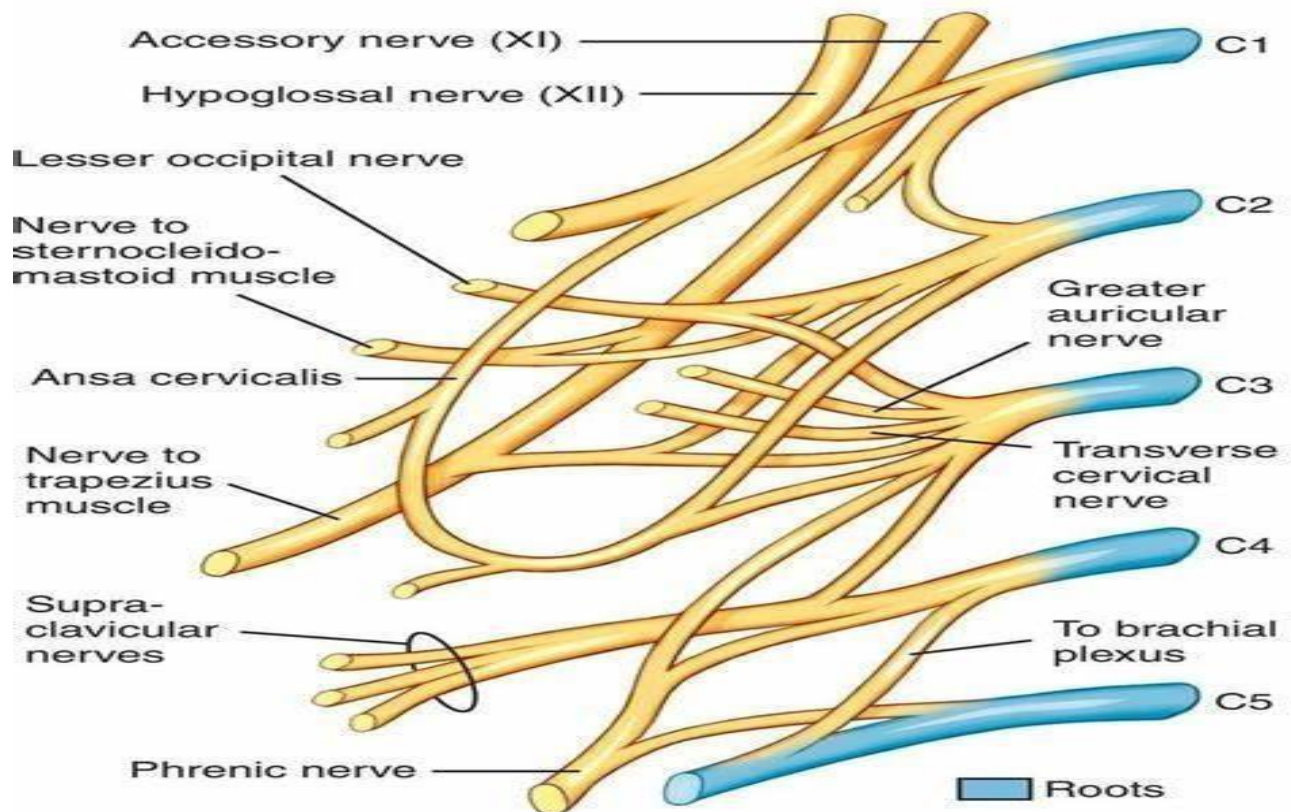


Lecture (4): Nerves of the

head & neck II

Vestibulocochlear nerve (CN VIII)

- **Cranial nerve 8** is a special somatic afferent nerve.
- It is comprised of **two parts: vestibular nerve and cochlear nerve.**



Lecture (5): Blood supply & lymphatic drainage of the head & neck

□ Arteries:

Common carotid

artery:

▪ Begin:

The **left common carotid artery** arises from the arch of the aorta. The **right common carotid artery** arises from brachiocephalic artery.

- **Course:** It ascends inside the carotid sheath **lateral** to the trachea and esophagus.
- **End:** at the upper border of thyroid cartilage opposite **C4**, where it splits into its two **terminal branches: the external and internal carotid arteries.**
- The **carotid body** and the **carotid sinus** are two specialized sensory organs associated with the bifurcation of the common carotid artery and its internal branch.